

The roles of fuzzy logic and soft computing in the conception, design and deployment of intelligent systems

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The essence of soft computing is that, unlike the traditional, hard computing, it is aimed at an accommodation with the pervasive imprecision of the real world. Thus, the guiding principle of soft computing is: ‘...exploit the tolerance for imprecision, uncertainty and partial truth to achieve tractability, robustness, low solution cost and better rapport with reality’. In the final analysis, the role model for soft computing is the human mind.

Soft computing is not a single methodology. Rather, it is a partnership. The principal partners at this juncture are fuzzy logic, neuro-computing and probabilistic reasoning, with the latter subsuming genetic algorithms, chaotic systems, belief networks and parts of learning theory.

In coming years, the ubiquity of intelligent systems is certain to have a profound impact on the ways in which man-made intelligent systems are conceived, designed, manufactured, employed and interacted with. It is within this perspective that the basic issues relating to soft computing and intelligent systems are addressed in this paper.

1. Introduction

To see the evolution of fuzzy logic in a proper perspective, it is important to note that we are in the throes of what is popularly called the information revolution. The artefacts of this revolution are visible to all. The Internet, World Wide Web, cellular phones, facsimile machines and home computers with powerful information processing capabilities have all become a part of everyday reality. The centrality of information in almost everything that we do is a fact that few would care to challenge.

Much less visible, but potentially of equal or even greater importance, is what might be called the intelligent systems revolution. The artefacts of this revolution are man-made systems which exhibit an ability to reason, learn from experience and make rational decisions without human intervention. I coined the term MIQ (machine intelligence quotient) to describe a measure of intelligence of man-made systems. In this perspective, an intelligent system is a system which has a high MIQ.

I will have more to say about MIQ at a later point. A question that I should like to raise now is the following. We have been talking about artificial intelligence for over four decades. Why did it take AI so long to yield visible results?

Let me cite an example that bears on this question. When I was an instructor at Columbia University I wrote a

paper entitled ‘Thinking Machines — A New Field in Electrical Engineering’ which was published in a student magazine [1]. In the opening paragraph of that article, I quoted a number of headlines which appeared in the popular press of the time. One of the headlines read: ‘An Electric Brain Capable of Translating Foreign Languages is Being Built’. The point is that my article was published in January 1950, about six years before the term ‘artificial intelligence’ was coined. What is obvious today is that a translation machine could not have been built in 1950 or earlier. The requisite technologies and methodologies were not in place.

We are much more humble today than we were at that time. The difficulty of building systems that could mimic human reasoning and cognitive ability turned out to be much greater than we thought at that time. Even today, with a vast array of powerful tools at our disposal, we are still incapable of building machines that can do what many children can do with ease, e.g. understand a fairy tale, peel an orange, or eat food with a knife and a fork.

At this point, let me return to the concept of MIQ. A basic difference between IQ and MIQ is that IQ is more or less constant whereas MIQ changes with time and is machine-specific. Furthermore, the dimensions of MIQ and IQ are not the same. For example, speech recognition might

be an important dimension of MIQ but in the case of IQ, it is taken for granted.

At this juncture, we do not have as yet an agreed set of tests to measure the MIQ of a man-made system, e.g. a camcorder; but I believe that such tests will be devised at some point in the future and that eventually the concept of MIQ will play an important role in defining and measuring machine intelligence.

In realistic terms, we are just beginning to enter the age of intelligent systems. Why did it take so long for this to happen?

In my perception, there are three main reasons. In the first place, until recently the principal tools in AI's armamentarium were centred on symbol manipulation and predicate logic, while the use of numerical techniques was looked upon with disfavour. What is more obvious today than it was in the past is that symbol manipulation and predicate logic have serious limitations in dealing with real world problems in the realms of computer vision, speech recognition, handwriting recognition, image understanding, multimedia database search, motion planning, common-sense reasoning and in the management of uncertainty.

2. Soft computing and fuzzy logic

During the past several years, our ability to conceive, design and build machines with high MIQ has been greatly enhanced by the advent of what is now referred to as soft computing. Soft computing (SC) is not a single methodology. Rather, it is a consortium of computing methodologies which collectively provide a foundation for the conception, design and deployment of intelligent systems. At this juncture, the principal members of SC are fuzzy logic (FL), neuro-computing (NC), genetic computing (GC), and probabilistic reasoning (PR), with the last subsuming evidential reasoning, belief networks, chaotic systems, and parts of machine learning theory. In contrast to the traditional hard computing, soft computing is tolerant of imprecision, uncertainty and partial truth. The guiding principle of soft computing is: '...exploit the tolerance for imprecision, uncertainty and partial truth to achieve tractability, robustness, low solution cost and better rapport with reality'.

What is important about soft computing is that its constituent methodologies are for the most part synergistic and complementary rather than competitive. Thus, in many cases, higher MIQ can be achieved by employing FL, NC, GC and PR in combination rather than singly. Furthermore, there are many problems which cannot be solved if the only tool is fuzzy logic, neuro-computing, genetic computing or probabilistic reasoning. This challenges the position of those who claim that their favourite tool, be it FL, NC, GC or PR, is capable of solving all problems. The proponents of

such views will certainly shrink in number once a better understanding of soft computing becomes widespread.

Within SC, each of the constituent methodologies has a set of capabilities to offer. In the case of fuzzy logic, it is a body of concepts and techniques for dealing with imprecision, information granularity, approximate reasoning and, most importantly, computing with words. In the case of neuro-computing, it is the capability for learning, adaptation and identification. In the case of genetic computing, it is the capability to employ systematised random search and achieve optimal performance. And in the case of probabilistic reasoning, it is a body of concepts and techniques for uncertainty management and evidential reasoning.

Systems in which FL, NC, GC and PR are used in some combination are called hybrid systems. Among the most visible systems of this type are the so-called neuro-fuzzy systems. We are beginning to see fuzzy-genetic systems, neuro-genetic systems and neuro-fuzzy-genetic systems. In my view, eventually, most high-MIQ systems will be hybrid systems. In the future, the ubiquity of hybrid systems will have a profound impact on the ways in which intelligent systems are designed, built and interacted with.

What is the place of fuzzy logic in soft computing? First, I should like to clarify a common misconception about what fuzzy logic is and what it has to offer.

A source of confusion is that the label fuzzy logic is used in two different senses. In a narrow sense, fuzzy logic is a logical system which is an extension of multi-valued logic. However, even in its narrow sense the agenda of fuzzy logic is very different both in spirit and in substance from the agendas of multi-valued logical systems.

In its wide sense — which is the sense in predominant use today — fuzzy logic is coextensive with the theory of fuzzy sets, that is, classes with unsharp boundaries [2]. In this perspective, fuzzy logic in its narrow sense is a branch of fuzzy logic in its wide sense.

What is important about FL is that any theory, T, can be fuzzified — and hence generalised — by replacing the concept of a crisp set in T with that of a fuzzy set. In this way, one is led to a fuzzy T, e.g. fuzzy arithmetic, fuzzy topology, fuzzy probability theory, fuzzy control and fuzzy decision analysis. What is gained from fuzzification is greater generality and better rapport with reality. However, fuzzy numbers are more difficult to compute with than crisp numbers. Furthermore, the meanings of most fuzzy concepts are context- and/or application-dependent. This is the price that has to be paid for a better rapport with reality.

3. Information granularity

There is a point of fundamental importance which lies at the base of ways in which humans deal with fuzzy concepts. The point in question has to do with information granularity and its role in human reasoning, communication and concept formation. In what follows, I will attempt to explain why information granularity plays an essential role in dealing with fuzzy concepts and, in particular, in reasoning and computing with words rather than numbers.

The concept of information granularity motivated most of my early work on fuzzy sets and fuzzy logic. Basically, the point that I stressed is that most human concepts are fuzzy because they are the result of clumping of points or objects which are drawn together by similarity. The fuzziness of such clumps, then, is a direct consequence of the fuzziness of the concept of similarity. Simple examples of clumps are the concepts of 'middle-aged', 'downtown', 'partially cloudy', 'obtuse', etc. To underscore its role, a clump will be referred to as a granule.

In a natural language, words play the role of labels of granules. In this role, words serve to achieve data compression. The achievement of data compression through the use of words is a key facet of human reasoning and concept formation.

In fuzzy logic, information granularity underlies the concepts of linguistic variable and fuzzy if-then rules [3, 4]. These concepts were formally introduced in my 1973 paper 'Outline of a New Approach to the Analysis of Complete Systems and Decision Processes' [5]. Today, almost all applications of fuzzy logic employ these concepts. It is of historical interest to note that my introduction of these concepts was met with scepticism and hostility by many eminent members of the scientific establishment.

The importance of fuzzy rules stems from the fact that such rules are close to human intuition. In fuzzy logic, fuzzy rules play a central role in what is called Fuzzy Dependency and Command Language (FDCL). In an informal way, it is this language that is used in most of the applications of fuzzy logic.

In comparing fuzzy logic with other methodologies, a point that is frequently unrecognised is that, typically, the point of departure in a fuzzy logic solution is a human solution. Thus, a fuzzy logic solution is usually a human solution expressed in FDCL. An easily understood example of this point is the car parking problem in which the objective is to place the car near the curb and almost parallel to it. A fuzzy logic solution of the parking problem would be a collection of fuzzy if-then rules which describe how a human parks a car. The parking problem is hard to solve in the context of classical control. In this context, the point of departure is not a human solution but a description of the

final state, the initial state, the constraints and the equations of motion.

A further example which illustrates the essentiality of information granularity is the following. Consider a situation in which a person A is talking over the phone to a person B whom A does not know. After a short time, say 10-20 seconds, A can form a rough estimate of the age of B expressed as:

the probability that B is very young is very low
the probability that B is young is low
the probability that B is middle-aged is high
the probability that B is old is low
the probability that B is very old is very low

These estimates may be interpreted as a granular representation of the probability distribution, P, of B's age. In a symbolic form, P may be represented as a fuzzy graph:

$$P = \text{very low} \setminus \text{very young} + \text{low} \setminus \text{young} + \text{high} \setminus \text{middle-aged} \\ + \text{low} \setminus \text{old} + \text{very low} \setminus \text{very old}$$

In this expression, '+' is the disjunction operator and a term such as low \setminus old means that low is the linguistic probability that B is old.

The important point is that humans can form such estimates using linguistic, i.e. granulated, values of age and probabilities. However, humans could not come up with numerical estimates of the form 'the probability that B is 25 is 0.012.'

It should be observed that in many cases a human would estimate the age of B as middle-aged omitting the associated probability. The omission of probabilities may be justified if there exists what might be called a p-dominant value in a probability distribution, i.e. a value whose probability dominates the probabilities of other values. The omission of probabilities plays a key role in approximate reasoning [6].

A question which arises is: 'Could the use of a methodology within soft computing provide an estimate of the age of B without human intervention?' In my view, the answer is in the negative. More specifically, neuro-computing and genetic computing techniques would fail because of the complexity of input/output pairs, while fuzzy logic would fail — even though a human solution exists — because humans would not be able to articulate the rules by which the age estimate is arrived at.

In sum, information granularity lies at the centre of human reasoning, communication and concept formation. Within fuzzy logic, it plays a pivotal role in what might be called computing with words, or CW for short. CW may be viewed as one of the most important contributions of fuzzy logic. What is computing with words? As its name suggests,

in computing with words the objects of computing are words rather than numbers, with words playing the role of labels of granules. Very simple examples of CW are:

Dana is young and Tandy is a few years older than Dana,
 $\therefore$ Tandy is (young + few) years old

Most students are young and most young students are single,
 $\therefore$ most² students are single

In these examples, young, few and most are fuzzy numbers; + is the operation of addition in fuzzy arithmetic and most² is the square of most in fuzzy arithmetic.

In Western cultures, there is a deep-seated tradition of according more respect to numbers than to words; but, as is true of any tradition, a time comes when the rationale for a tradition ceases to be beyond question. In my view, the time has come for questioning the validity of the tradition of according more respect to numbers than to words.

What we need at this juncture is a system which allows the data to be expressed as propositions in a natural language. This is what CW attempts to provide. The point of departure in CW is a collection of propositions expressed in a natural language. This collection is referred to as the initial data set (IDS). The desired answers or conclusions are likewise expressed as a collection of propositions expressed in a natural language. This collection is referred to as the terminal data set (TDS). The problem is to arrive at TDS starting with IDS. A very simple example is one where the initial data set is the proposition, 'most Swedes are tall', and the terminal data set is the answer to the query, 'What is the average height of Swedes?' The answer is expected to be of the form, 'the average height of Swedes is A, where A is a linguistic value of height'. In this example, the aim of CW is to compute A from the information provided by the initial data set.

In CW, words play the role of fuzzy constraints and a proposition is interpreted as a fuzzy constraint on a variable. For example, the proposition 'Mary is young' is interpreted as a fuzzy constraint on Mary's age. In symbols:

Mary is young $\rightarrow$ Age(Mary) is young

In this expression, ' $\rightarrow$ ' represents the operation of explicitation; 'Age (Mary)' is the constrained variable; and 'young' is a fuzzy relation which constrains 'Age(Mary)'.

More generally, if p is a proposition in a natural language, the result of explicitation of p is what is called the canonical form of p. Basically, the canonical form of a proposition, p, makes explicit the implicit fuzzy constraint in p, and thus serves to define the meaning of p as a constraint on a variable. In a more general setting, the canonical form of p is represented as:

$X \text{ is } R$

where X is the linguistic constrained variable, e.g. Age(Mary), R is the constraining fuzzy relation, e.g. young, and *isr* is a variable in which *r* is a discrete variable whose values define the role of R in relation to X. In particular, if $r = d$, *isd* is abbreviated to 'is' and the constraint 'X is R' is said to be disjunctive. In this case, R defines the possibility distribution of X. What is the reason for treating *r* as a variable? The richness of natural languages necessitates the use of a wide variety of constraints to represent the meaning of a proposition expressed in natural language. In CW, the principal types of constraints that are employed — in addition to the disjunctive type — are: conjunctive, probabilistic, usuality, random set, rough set, fuzzy graph and functional types. Each of these types corresponds to a particular value of *r*.

In CW, the first step in computing the terminal data set is that of explicitation, i.e. the representation of propositions in IDS in their canonical forms. The second step involves constraint propagation, which is carried out through the use of the rules of inference in fuzzy logic. In effect, the rules of inference in fuzzy logic may be interpreted as the rules of constraint propagation.

The third and final step in the computation of the terminal data set involves a retranslation of induced constraints into propositions expressed in a natural language. In fuzzy logic, this requires the use of what is referred to as linguistic approximation.

What is important to recognise is that the steps sketched above may require an extensive use of computing with numbers. However, as a stage of CW, computing with numbers takes place behind a curtain, hidden from the view of a user.

So what is it that CW has to offer? The ability to infer from initial data sets in which information is conveyed by propositions expressed in a natural language, opens the door to the formulation and solution of many problems in which the available information is not precise enough to justify the use of conventional techniques. To illustrate, suppose that the problem is that of maximising a function which is described in words through the fuzzy if-then rules:

if X is small then Y is small

if X is medium then Y is large

if X is large then Y is small

in which small, medium and large are defined through their membership functions. Another problem in this vein is the following. Assume that a box contains ten balls of various

sizes of which several are large and a few are small. What is the probability that a ball drawn at random is neither small nor large?

In these examples, the propositions in the initial data set are quite simple. The real challenge is to develop CW to a point where it could cope with propositions of much greater complexity which express real-world knowledge.

At this juncture, computing with words is a branch of fuzzy logic. In my perception, in coming years computing with words is likely to evolve into an important methodology in its own right, providing a way of coping with the pervasive imprecision and uncertainty of the real world [6]. In this perspective, the role model for computing with words, fuzzy logic and soft computing is the human mind.

4. Conclusions

The conception, design and deployment of intelligent systems presents a great challenge to those of us who are engaged in the development and applications of fuzzy logic and soft computing. Hopefully, our efforts will contribute to the creation of a society in which intelligent systems will serve to enhance human welfare and intellectual freedom.

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